

Sea Water Pressure Signal Malfunction

Symptoms

- The original equipment manufacturer (OEM) sea water pressure sensor has malfunctioned.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The sea water pressure sensor is connected to the alarm and safety C2 connector located on the customer interface box (C.I.B.).

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the C.I.B. wiring.	
	STEP 1A. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for an open circuit.	Greater than 100k ohms?
	STEP 1B. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a wire-to-wire short circuit.	Less than 10 ohms?
	STEP 1C. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a short circuit to ground.	Less than 10 ohms?
STEP 2.	Check the OEM wiring harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for an open circuit.	Greater than 100k ohms?
	STEP 2B. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a wire-to-wire short circuit.	Less than 10 ohms?
	STEP 2C. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a short circuit to ground.	Less than 10 ohms?
	STEP 2D. Check the sea water pressure SUPPLY +24-VDC wire for voltage.	+24-VDC?

STEP 1. Check the C.I.B. wiring.

STEP 1A. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for an open circuit.

Conditions : • Open the C.I.B. • Disconnect the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires at the control panel.		
Action	Specification/Repair	Next Step

Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for an open circuit. Note: An alarm will sound on the remote input/output unit when an open circuit is detected. - Place one test lead on the sea water pressure SIGNAL wire at the control panel. - Place the other test lead on the sea water pressure SIGNAL pin at the C2 connector. - Place one test lead on the sensor SUPPLY +24-VDC wire at the control panel. - Place the other test lead on the sensor SUPPLY +24-VDC pin at the C2 connector. Refer to the circuit diagram or wiring diagram for connector pin identification.	Greater than 100k ohms? YES	1B
	Greater than 100k ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 1B. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a wire-to-wire short circuit.

Conditions : - Open the C.I.B. - Disconnect the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires at the control panel. - Disconnect the C2 connector.		
Action	Specification/Repair	Next Step

Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a wire-to-wire short circuit. • Place one test lead on the sea water pressure SIGNAL wire at the control panel. • Place the other test lead on all other wires at the control panel. • Place one test lead on the sensor SUPPLY +24-VDC wire at the control panel. • Place the other test lead on all other wires at the control panel. Refer to the circuit diagram or wiring diagram for connector pin identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Less than 10 ohms? NO	1C

STEP 1C. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a short circuit to ground.

Conditions : • Open the C.I.B. • Disconnect the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires at the control panel. • Disconnect the C2 connector.		
Action	Specification/Repair	Next Step

Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a short circuit to ground. - Place one test lead on the sea water pressure SIGNAL wire at the control panel. - Place the other test lead on the panel ground. - Place one test lead on the sensor SUPPLY +24-VDC wire at the control panel. - Place the other test lead on the panel ground. Refer to the circuit diagram or wiring diagram for connector pin identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check the OEM wiring harness.

STEP 2A. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for an open circuit.

Conditions :

- Disconnect the OEM harness at the C2 and C9 connectors.
- Disconnect the sea water pressure sensor.

Action	Specification/Repair	Next Step

Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for an open circuit.

Note: An alarm will sound on the remote input/output unit when if a false alarm has occurred.

Less than 10 ohms?

YES

2B

- Place one test lead on the sea water pressure SIGNAL pin at the C2 connector.
- Place the other test lead on the sea water pressure SIGNAL pin at the C9 connector.
- Place one test lead on the sea water pressure sensor SUPPLY +24-VDC pin at the C2 connector.
- Place the other test lead on the sea water pressure sensor SUPPLY +24-VDC pin at the C9 connector.
- Place one test lead on the sea water pressure SIGNAL pin at the C9 connector.
- Place the other test lead on the sea water pressure SIGNAL pin at the sensor connector.
- Place one test lead on the sea water pressure sensor SUPPLY +24-VDC pin at the C9 connector.
- Place the other test lead on the sea water

pressure sensor
SUPPLY +24-VDC pin at
the sensor connector.

Refer to the circuit diagram
or wiring diagram for
connector pin identification.

Less than 10 ohms?

NO

Repair :

Replace the wire. Refer to
Procedure 015-023 in
Section 15.
(/qs3/pubsys2/xml/en/proc
edures/300/300-015-
023.html)

Repair complete

**STEP 2B. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires
for a wire-to-wire short circuit.**

Conditions :

- Disconnect the OEM harness at the C2 and C9 connectors.
- Disconnect the sea water pressure sensor.

Action	Specification/Repair	Next Step
Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a wire-to-wire short circuit. • Place one test lead on the sea water pressure SIGNAL pin at the C2 connector. • Place the other test lead on all other pins at the C2 connector. • Place one test lead on the sensor SUPPLY +24-VDC pin at the C2 connector. • Place the other test lead on all other pins at the C2 connector. • Place one test lead on the sea water pressure SIGNAL pin at the C9 connector. • Place the other test lead on all other pins at the C9 connector. • Place one test lead on the sensor SUPPLY +24-VDC pin at the C9 connector.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
• Place the other test lead on all other pins at the C9 connector. Refer to the circuit diagram or wiring diagram for connector pin identification.	Less than 10 ohms? NO	2C

STEP 2C. Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a short circuit to ground.

Conditions :

- Disconnect the OEM harness at the C2 and C9 connectors.
- Disconnect the sea water pressure sensor.

Action	Specification/Repair	Next Step
Check the sea water pressure SIGNAL and sensor SUPPLY +24-VDC wires for a short circuit to ground. • Place one test lead on the sea water pressure SIGNAL pin at the C2 connector. • Place the other test lead on the engine ground. • Place one test lead on the sensor SUPPLY +24-VDC pin at the C9 connector. • Place the other test lead on the engine ground.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
Refer to the circuit diagram or wiring diagram for connector pin identification.	Less than 10 ohms? NO	2D

STEP 2D. Check the sea water pressure SUPPLY +24-VDC wire for voltage.

Conditions :

- Disconnect the sea water pressure sensor connector.

Action	Specification/Repair	Next Step
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Check the sea water pressure SUPPLY +24-VDC wire for voltage. - Place one test lead on the sea water pressure sensor SUPPLY +24-VDC pin at the sensor connector. - Place the other test lead on the engine ground. Refer to the circuit diagram or wiring diagram for connector pin identification.	+24-VDC? YES Repair : Replace the sea water pressure sensor. Refer to the OEM service manual. Refer to Procedure 019-452 in Section 19. (/qs3/pubsys2/xml/en/procedures/300/300-019-452.html)	Repair complete
	+24-VDC? NO Repair : Replace the remote input/output unit. Contact a Cummins® Authorized Repair Location.	Repair complete